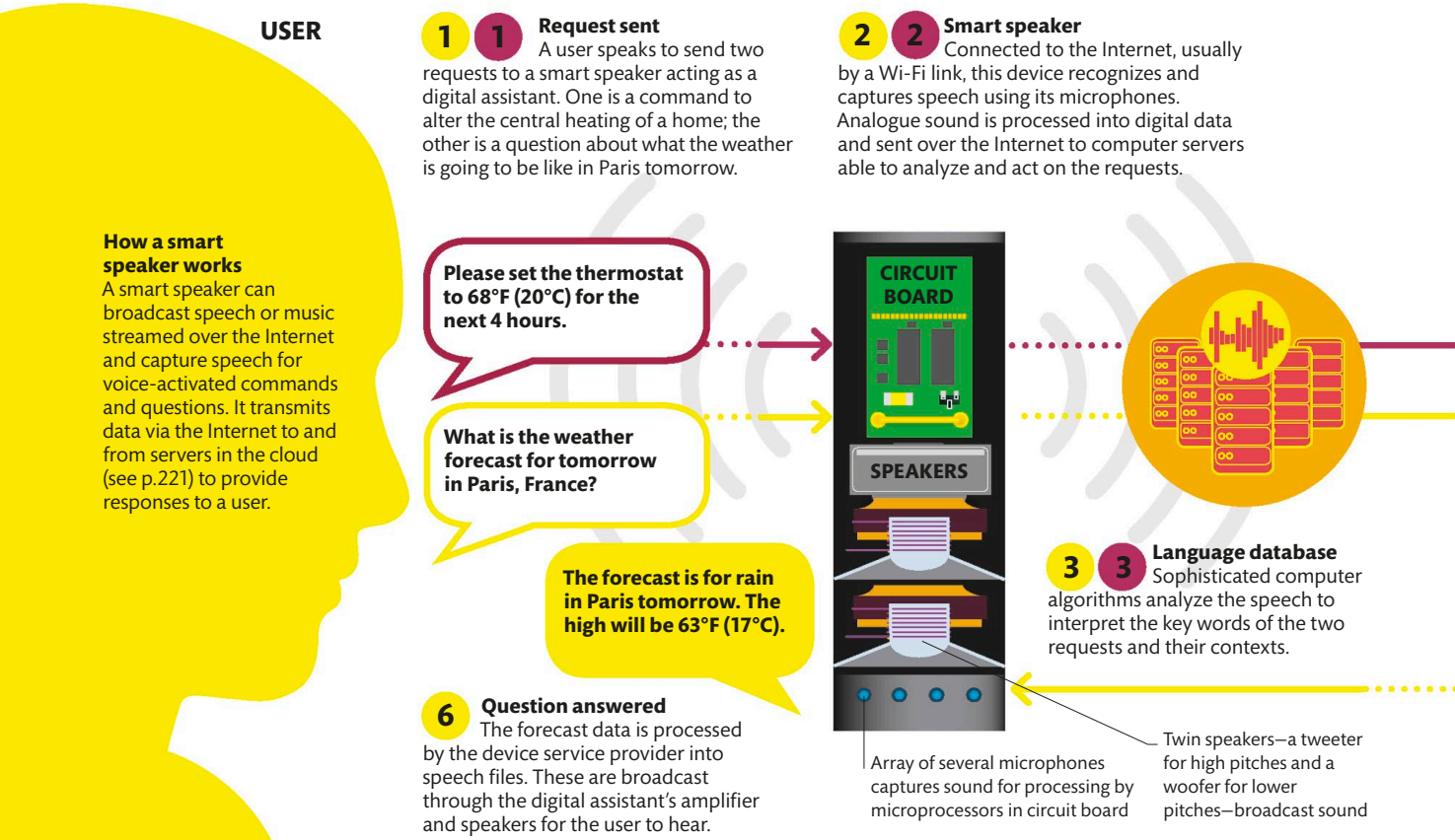
Digital assistants

These versatile devices exist as apps on smartphones and as household hardware such as smart speakers. They use voice-recognition algorithms to understand a user's commands and questions. They then direct these requests over the Internet to, for example, run an entertainment application or access an information service.

TO MAKE THEM SOUND MORE HUMAN, SOME DIGITAL ASSISTANTS ARE PROGRAMMED TO INSERT PAUSES INTO THEIR SENTENCES

How a smart speaker works
A smart speaker can broadcast speech or music streamed over the Internet and capture speech for voice-activated commands and questions. It transmits data via the Internet to and from servers in the cloud (see p.221) to provide responses to a user.



WHAT WAS THE FIRST SMART HOME DEVICE?

In 1966, US engineer Jim Sutherland built the Echo IV smart home computer system, which was capable of controlling lighting, heating, and TVs.

The digital home

The rapid rise in computing power, the Internet, and embedded microprocessors in everyday devices allows millions of devices to be connected and controlled over computer networks. As more and more connected devices enter households, technology enables people to control many household tasks while they are away from home, such as adjusting the central heating thermostat via a smartphone app.